

SCJP 1.6 (CX-310-065 , CX-310-066)

Subject: Mock Questions Set-4

Total Questions : 60

Prepared by : <http://www.javacertifications.net>

SCJP 6.0: Full Length Mock Exam Set-4(60 Questions)

Questions Question - 1

What is the output?

```
class Bird {
    boolean b;

    public Bird() {
        System.out.println(b);
    }
    public static void main(String[] args){
        Bird b = new Bird();
    }
}
```

- 1.true
- 2.false
- 3.garbage
- 4.none of the above

Explanation :

B is the correct answer.

default value of boolean is false.

Question - 2

How does the set collection deal with duplicate elements?

- 1.An exception is thrown if you attempt to add an element with a duplicate value
- 2.The add method returns false if you attempt to add an element with a duplicate value
- 3.A set may contain elements that return duplicate values from a call to the equals method
- 4.Duplicate values will cause an error at compile time

Explanation :

B is the correct answer.

add method of Set returns false if you attempt to add an element with a duplicate value.

Question - 3

```
if(0.0 == -0.0) {  
    System.out.println("true");  
}  
else{  
    System.out.println("false");  
}
```

1. prints false
2. prints true
3. none of the above
4. none of the above

Explanation :

B is the correct answer.

0.0 and -0.0 are same.

Question - 4

Which of the following are methods of the Thread class?

- 1) yield()
- 2) sleep(long msec)
- 3) go()
- 4) stop()

1. 1,3,4
2. 1,2,4
3. 1,3
4. 3

Explanation :

B is the correct answer.

go() is not related to Thread class.

Question - 5

What will be the result of executing the following code?

```
public class Test {  
  
    public static void main(String argv[]){  
        boolean a = true;  
        boolean b = false;  
        boolean c = true;  
        if (a == true)
```

```

        if (b == true)
        if (c == true)      System.out.println("c is true");
        else                System.out.println("nothing
is true");
        else if (a && (b = c)) System.out.println("confusing to
tell what is true and what is false");
        else
System.out.println("compile error");
    }
}

```

- 1.The code won't compile
- 2."c is true" will be printed
- 3."nothing is true" will be printed
- 4."confusing to tell what is true and what is false" will be printed

Explanation :

D is the correct answer.

"confusing to tell what is true and what is false" will be printed

Question - 6

```
public static double sin(double angle)
```

What are the units of the "angle" argument?

- 1.Degrees
- 2.Radians
- 3.Both
- 4.None of the above

Explanation :

B is the correct answer.

No Explanation Available

Question - 7

A signed data type has an equal number of non-zero positive and negative values available

- 1.true
- 2.false
- 3.none of the above
- 4.none of the above

Explanation :

B is the correct answer.

The range of negative numbers is greater by 1 than the range of positive numbers

Question - 8

What is output of the below code ?

```
public class A {
    public A() {
        System.out.println("A");
    }
}

public class B extends A implements Serializable {
    public B() {
        System.out.println("B");
    }
}

public class Test {

    public static void main(String... args) throws Exception {
        B b = new B();

        ObjectOutputStream save = new ObjectOutputStream(new
        FileOutputStream("datafile"));
        save.writeObject(b);
        save.flush();

        ObjectInputStream restore = new ObjectInputStream(new
        FileInputStream("datafile"));
        B z = (B) restore.readObject();

    }
}
```

- 1.A B A
- 2.A B A B
- 3.B B
- 4.B

Explanation :

A is the correct answer.

On the time of deserialization , the Serializable object not create new object. So constructor of class B does not called. A is not Serializable object so constructor is called.

Question - 9

What is output of the below code ?

```
public class A {}

public class B implements Serializable {
    A a = new A();
    public static void main(String... args){
        B b = new B();
        try{
            FileOutputStream fs = new FileOutputStream("b.ser");
            ObjectOutputStream os = new ObjectOutputStream(fs);
            os.writeObject(b);
            os.close();

        }catch(Exception e){
            e.printStackTrace();
        }

    }
}
```

- 1.Compilation Fail
- 2.java.io.NotSerializableException: Because class A is not Serializable.
- 3.No Exception
- 4.None of the above

Explanation :

B is the correct answer.

java.io.NotSerializableException:A Because class A is not Serializable.

Question - 10

What is the output?

```
public class Test {

    public static void main(String[] args) {

        StringBuffer sb = new StringBuffer("ssss");
        String st = new String("ssss");
        System.out.println("st equals sb : " + st.equals(sb));

    }

}
```

- 1.st equals sb : false
- 2.st equals sb : true
- 3.Compile Error
- 4.Runtime Exception

Explanation :

A is the correct answer.

String's equals() method checks if the argument is of type string, if not it returns false

Question - 11

Which statement is true?

- 1.an inner class instance can never stand alone without a direct relationship to an instance of the outer class
- 2.you can access the inner class is through a live instance of the outer class
- 3.Inner class can access private member of the outer class.
- 4.All of the above

Explanation :

D is the correct answer.

an inner class instance can never stand alone without a direct relationship to an instance of the outer class. you can access the inner class is through a live instance of the outer class. Inner class can access private member of the outer class.

Question - 12

What is output of the below code ?

```
public class Outer {
    private int a = 7;

    class Inner {
        public void displayValue() {
            System.out.println("Value of a is " + a);
        }
    }
}

public class Test {

    public static void main(String... args) throws Exception {
        Outer mo = new Outer();
        Outer.Inner inner = mo.new Inner();
        inner.displayValue();
    }
}
```

}

- 1.Value of a is 7
- 2.Compile Fail
- 3.Runtime Exception
- 4.None of the above

Explanation :

A is the correct answer.

an inner class instance can never stand alone without a direct relationship to an instance of the outer class. you can access the inner class is through a live instance of the outer class. Inner class can access private member of the outer class.

Question - 13

Which of the following return true?

- 1."das" == new String("das")
- 2."das".equals("das")
- 3."das".equals(new Button("das"))
- 4.None of the above

Explanation :

A is the correct answer.

== compare the addresses and equals() methods checks for content. In the "das" == new String("das") , new String("das") point to different address then "das".

Question - 14

Which of the following is correct?

- 1.String temp [] = new String {"j" "a" "z"};
- 2.String temp [] = { "j " " b" "c"};
- 3.String temp = {"a", "b", "c"};
- 4.String temp [] = {"a", "b", "c"};

Explanation :

D is the correct answer.

String temp [] = {"a", "b", "c"}; where temp[] is String array.

Question - 15

What is the correct declaration of an abstract method that is intended to be public?

1. public abstract void add();
2. public abstract void add() {}
3. public abstract add();
4. public virtual add();

Explanation :

A is the correct answer.

abstract method should not have body.

Question - 16

A String Class

1. final
2. non final
3. implements Serializable interface
4. abstract class

Explanation :

A and C is the correct answer.

Strings are constant; their values cannot be changed after they are created.

Question - 17

Under what situations do you obtain a default constructor?

1. When you define any class
2. When the class has no other constructors
3. When you define at least one constructor
4. None of the above

Explanation :

B is the correct answer.

When a class has no other constructors, default will be automatically there.

Question - 18

Which of the following can be used to define a constructor for this class, given the following code:

```
public class Test {  
    ...  
}
```

1. public void Test() {...}
2. public Test() {...}
3. public static Test() {...}
4. public static void Test() {...}

Explanation :

B is the correct answer.

Constructor should not have any return type and should not be static.

Question - 19

Assuming a method contains code which may raise an **Exception** (but not a **RuntimeException**), what is the correct way for a method to indicate that it expects the caller to handle that exception?

1. throw Exception
2. throws Exception
3. new Exception
4. Don't need to specify anything

Explanation :

B is the correct answer.

throws Exception from method and the caller method should catch to handle the exception.

Question - 20

What is the result of executing the following code, using the parameters 3 and 0:

```
public void divide(int a, int b) {  
    try {  
        int c = a / b;  
    } catch (Exception e) {  
        System.out.print("Exception ");  
    } finally {  
        System.out.println("Finally");  
    }  
}
```

1. Prints out: Exception Finally
2. Prints out: Finally
3. Prints out: Exception

4. None of the above.

Explanation :

A is the correct answer.

finally block always executed whether exception occurs or not. 3/0 throw exception.

Question - 21

What is the result of executing the following code, using the parameters 0 and 3 ?

```
public void divide(int a, int b) {  
    try {  
        int c = a / b;  
    } catch (Exception e) {  
        System.out.print("Exception ");  
    } finally {  
        System.out.println("Finally");  
    }  
}
```

1. Prints out: Exception Finally
2. Prints out: Finally
3. Prints out: Exception
4. None of the above.

Explanation :

B is the correct answer.

finally block always executed whether exception occurs or not. 0/3 = 0

Question - 22

Which of the following is a legal return type of a method overloading the following method ?

```
public void add(int a) {...}
```

1. void
2. int
3. String
4. Can be anything

Explanation :

D is the correct answer.

method overloading don't care about return type.

Question - 23

Which of the following statements is correct for a method which is overriding the following method ?

```
public void add(int a) {...}
```

- 1.the overriding method must return void
- 2.the overriding method must return int
- 3.the overriding method may return anything
- 4.None of the above

Explanation :

A is the correct answer.

In case of overriding , super class method and the sub class method should have same return type (jre 1.4). In jre 1.5 and jre 1.6 - subclass method can have same return type or subclass of the return type.

Question - 24

What is the value of y?

```
int y = 2 % 4;
```

- 1.0
- 2.2
- 3.4
- 4.Compile Error

Explanation :

B is the correct answer.

$x \% y$ is x if x is less than y.

Question - 25

Is this code legal ?

```
public class Test {  
    void aMethod(){  
        static int b = 10;  
        System.out.println(b);  
    }  
}
```

- 1.Yes
- 2.No
- 3.Can't Say

4. None of the above

Explanation :

B is the correct answer.

Local variables cannot be declared static.

Question - 26

```
class A {  
    B b = new B();  
    C c = (C) b;  
}
```

Referring to the above, when is the cast of "b" to class C allowed?

1. B and C are subclasses of the same superclass.
2. If B and C are super classes of the same subclass
3. B is a subclass of C.
4. B is a superclass of C.

Explanation :

C is the correct answer.

For Example. B is superclass and C is subclass then B b = new (B)C is allowed.

Question - 27

Is this legal

```
public class Test {  
    static { int a = 5; }  
    public static void main(String[] args) {  
        System.out.println(a);  
    }  
}
```

1. Yes
2. No
3. Can't Say
4. None

Explanation :

B is the correct answer.

A variable declared in a static initialiser is not accessible outside its enclosing block.

Question - 28

```
try{
    File f = new File("a.txt");
    }catch(Exception e){

    }catch(IOException io){

    }
}
```

Is this code create new file name a.txt ?

- 1.True
- 2.False
- 3.Compilation Error
- 4.None

Explanation :

C is the correct answer.

IOException is unreachable to compiler because all exception is going to catch by Exception block.

Question - 29

Which variables can an inner class access from the class which encapsulates it?

- 1.All static variables.
- 2.All final variables.
- 3.All instance variables
- 4.All of the above

Explanation :

A is the correct answer.

inner class can access static , final and instance variables of outer class.

Question - 30

What is the output ?

```
public class Test {
    final int k;
    public static void main(String[] args) {
        System.out.println(k+1);
    }
}
```

- 1.0
- 2.1
- 3.Compile Error.
- 4.None of the above.

Explanation :

C is the correct answer.

final variables should be initialized.

Question - 31

What class must an inner class extend ?

- 1.The top level class
- 2.The Object class
- 3.Any class or interface
- 4.It must extend an interface

Explanation :

C is the correct answer.

inner class can extends any class or interface.

Question - 32

Which one of the following is the equivalent of main() in a thread?

- 1.start()
- 2.run()
- 3.go()
- 4.begin()

Explanation :

B is the correct answer.

No Explanation Available.

Question - 33

What happens if a class defines a method with the same name and parameters as a method in its superclass?

- 1.The compiler automatically renames the subclass method.
- 2.The program runs since because overriding methods is allowed
- 3.The program throws a DuplicateMethod exception.
- 4.The compiler shows a DuplicateMethod error.

Explanation :
B is the correct answer.

Java supports Method Overriding.

Question - 34

What is the result of attempting to compile and run this code ?

```
class A extends Exception{}
class B extends A{}
class C extends B{}
public class Test {
    static void aMethod() throws C{ throw new C(); }
    public static void main(String[] args){
        int x = 10;
        try { aMethod(); }
        catch(A e) { System.out.println("Error A");}
        catch(B e) { System.out.println("Error B");}
    }
}
```

- 1.Compiler error
- 2.It will print "Error A"
- 3.It will print "Error B"
- 4.None of the above

Explanation :
A is the correct answer.

This will not compile because B is a subclass of A, so it must come before A. For multiple catch statements the rule is to place the the subclass exceptions before the superclass exceptions.

Question - 35

Which one of the following is a limitation of subclassing the Thread class?

- 1.You must catch the ThreadDeath exception.
- 2.You must implement the Threadable interface.
- 3.You cannot have any static methods in the class
- 4.You cannot subclass any other class.

Explanation :
D is the correct answer.

Java don't support multiple inheritance.Subclassing the Thread Class , you can't extend any other class.

Question - 36

What is the effect of issuing a wait() method on an object ?

- 1.If a notify() method has already been sent to that object then it has no effect
- 2.The object issuing the call to wait() will halt until another object sends a notify() or notifyAll() method
- 3.An exception will be raised
- 4.The object issuing the call to wait() will be automatically synchronized with any other objects using the receiving object.

Explanation :

B is the correct answer.

issuing the call to wait() will halt until another object sends a notify() or notifyAll() method.

Question - 37

What is the effect of adding the sixth element to a vector created in the following manner?

```
new Vector(5, 10);
```

- 1.An IndexOutOfBoundsException exception is raised.
- 2.The vector grows in size to a capacity of 10 elements
- 3.The vector grows in size to a capacity of 15 elements
- 4.Nothing, the vector will have grown when the fifth element was added

Explanation :

C is the correct answer.

Constructor of Vector public Vector(int initialCapacity, int capacityIncrement).

Adding the 6th element to a vector , Size will be initialCapacity+capacityIncrement , i.e $5+10 = 15$

Question - 38

Observe the code below

```
public class Test {
    int a[] = new int[4];
    void aMethod()
    {
        int b = 0 , index;
        a[a[b]] = a[b] = b = 2;
        index = ??? ;
        System.out.println(a[index]);
    }
}
```

```
}  
What value assigned to the variable index will print a non zero  
result?
```

- 1.0
- 2.1
- 3.2
- 4.3

Explanation :

A is the correct answer.

the operands are evaluated from left to right here so in the expression `a[a[b]]` the value of `b` is 0 and `a[0]` also holds the value 0 when the expression is evaluated so the array element at index 0 is assigned the value 2.

Question - 39

```
What is the output of the bellow code?  
Object a = "hello";  
String b = "hello";  
if(a == b)  
System.out.println("equal");  
else  
System.out.println("not equal");
```

- 1.equal
- 2.not equal
- 3.Code not compile
- 4.None of the above

Explanation :

A is the correct answer.

It does not matter if the object reference is of type `Object` this will still return true because both of them point to the same `String` object in the string pool.

Question - 40

```
Is the bellow statement True or False?  
The garbage collector is required to makes sure that all objects  
held by soft references are garbage collected before the VM  
throws  
an OutOfMemoryError.
```

- 1.True
- 2.False
- 3.Sometime true , sometime false.
- 4.None of the above.

Explanation :

A is the correct answer.

If the VM finds itself unable to allocate memory for a new object it is required to kick start the garbage collector and reclaim all objects that are not held by strong references before throwing an OutOfMemoryError.

Question - 41

```
public class A{
    private void test1(){

        System.out.println("test1 A");
    }
}

public class B extends A{
    public void test1(){

        System.out.println("test1 B");
    }
}

public class outputtest{
    public static void main(String[] args){
        A b = new B();
        b.test1();
    }
}
```

What is the output?

- 1.test1 A
- 2.test1 B
- 3.Not compile because test1() method in class A is not visible.
- 4.None of the above.

Explanation :

C is the correct answer.

Not compile because test1() method in class A is private so it is not visible.

Question - 42

```
public class A{
    private void test1(){

        System.out.println("test1 A");
    }
}
```

```

    }
}

public class B extends A{
public void test1(){

    System.out.println("test1 B");
}
}

public class Test{
public static void main(String[] args){
    B b = new B();
    b.test1();

}
}

```

What is the output?

- 1.test1 B
- 2.test1 A
- 3.Not compile because test1() method in class A is not visible
- 4.None of the above

Explanation :

A is the correct answer.

This is not related to superclass , B class object calls its own method so it compile and output normally.

Question - 43

```

public class A{
    public void test1(){

        System.out.println("test1 A");
    }
}

public class B extends A{
public void test1(){

    System.out.println("test1 B");
}
}

public class Test{
public static void main(String[] args){
A b = new B();
    b.test1();

}
}

```

What is the output?

- 1.test1 B
- 2.test1 A
- 3.Not compile because test1() method in class A is not visible
- 4.None of the above

Explanation :

A is the correct answer.

Superclass reference of subclass point to subclass method and superclass variables.

Question - 44

```
class c1
{
public static void main(String a[])
{
c1 ob1=new c1();
Object ob2=ob1;
System.out.println(ob2 instanceof Object);
System.out.println(ob2 instanceof c1);
}
}
```

- 1.Prints true,false
- 2.Print false,true
- 3.Prints true,true
- 4.compile time error

Explanation :

C is the correct answer.

instanceof operator checks for instance related to class or not.

Question - 45

```
class c2
{

{
System.out.println("initializer");
}

public static void main(String a[])
{
System.out.println("main");
c2 ob1=new c2();
}
```

}

1. prints main and initializer
2. prints initializer and main
3. compile time error
4. None of the above

Explanation :

A is the correct answer.

statement block executes on creation of object of the class.

Question - 46

1. `StringBuffer s1 = new StringBuffer("abc");`
2. `StringBuffer s2 = s1;`
3. `StringBuffer s3 = new StringBuffer("abc");`

How many objects are created ?

- 1.3
- 2.1
- 3.2
- 4.4

Explanation :

D is the correct answer.

s1 is one object, s2 is another object, s3 is another object and "abc" is another String Object .

Question - 47

```
class C
{
public static void main(String a[])
{
C c1=new C();
C c2=m1(c1);
C c3=new C();
c2=c3;           //6
anotherMethod();
}
static C m1(C ob1){
ob1 =new C();
return ob1;
}
}
```

After line 6, how many objects are eligible for garbage collection?

- 1.1
- 2.2
- 3.3
- 4.4

Explanation :

B is the correct answer.

No Explanation Available

Question - 48

```
class C{
    static int f1(int i) {
        System.out.print(i + ",");
        return 0;
    }
    public static void main (String[] args) {
        int i = 0;
        i = i++ + f1(i);
        System.out.print(i);
    }
}
```

- 1.Prints: 0,0
- 2.Prints: 1,0
- 3.Prints: 0,1
- 4.Compile-time error

Explanation :

B is the correct answer.

No Explanation Available

Question - 49

```
public class C {

    public C(){

    }

}

public class D extends C {

    public D(int i){

        System.out.println("tt");

    }

}
```

```

        public D(int i, int j){

            System.out.println("tt");

        }

    }

```

Is C c = new D(); works ?

- 1.It compile without any error
- 2.It compile with error
- 3.Can't say
- 4.None of the above

Explanation :

B is the correct answer.

C c = new D(); NOT COMPILE because D don't have default constructor. If super class has different constructor other then default then in the sub class you can't use default constructor

Question - 50

```

public class C {

    public C(){

    }

}

public class D extends C {

    public D(int i){

        System.out.println("tt");

    }

    public D(int i, int j){

        System.out.println("tt");

    }

}

```

Is C c = new D(8); works ?

- 1.It compile without any error
- 2.It compile with error
- 3.Can't say
- 4.None of the above

Explanation :

A is the correct answer.

C c = new D(8); COMPILE because D don't have default constructor. If super class has different constructor other then default then in the sub class you can't use default constructor

Question - 51

```
public class A {  
    { System.out.println("block");}  
    public A(){  
        System.out.println("A");  
    }  
  
    public static void main(String[] args){  
        A a = new A();  
    }  
}What will be output ?
```

- 1.A block
- 2.block A
- 3.A
- 4.None of the above

Explanation :

B is the correct answer.

First execute block then constructor.

Question - 52

```
public class A {  
    static{System.out.println("static");}  
    { System.out.println("block");}  
    public A(){  
        System.out.println("A");  
    }  
  
    public static void main(String[] args){  
        A a = new A();  
    }  
}What will be output ?
```

- 1.A block static
- 2.static block A
- 3.static A
- 4.A

Explanation :

B is the correct answer.

First execute static block, then block then constructor.

Question - 53

```
class Bird {
    { System.out.print("b1 "); }
    public Bird() { System.out.print("b2 "); }
}
class Raptor extends Bird {
    static { System.out.print("r1 "); }
    public Raptor() { System.out.print("r2 "); }
    { System.out.print("r3 "); }
    static { System.out.print("r4 "); }
}
class Hawk extends Raptor {
    public static void main(String[] args) {
        System.out.print("pre ");
        new Hawk();
        System.out.println("hawk ");
    }
}
```

What is the output?

- 1.r1 r4 pre b1 b2 r3 r2 hawk
- 2.r1 r4 pre b1 b2 hawk
- 3.r1 r4 pre b1 b2 hawk r3 r2
- 4.None of the above

Explanation :

A is the correct answer.

All static blocks execute first : sequence super class first
Blocks execute : sequence super class first

Question - 54

```
class Bird {
    int i;
    boolean b;
    float f;

    public Bird() {
        System.out.println(i);
        System.out.println(b);
        System.out.println(f);
    }
}
```

What is the output?

- 1.0 false 0.0
- 2.0 true 0.0

- 3. Not Compile
- 4. None of the above

Explanation :

A is the correct answer.

Jvm initialize default values; Static variables also;

Question - 55

```
public class Test {  
    public static void main(String[] args){  
        StringBuffer a = new StringBuffer("Hello");  
        StringBuffer b = a.append("World");  
        System.out.println(b);  
    }  
}
```

What is the output?

- 1. World
- 2. HelloWorld
- 3. World
- 4. None of the above

Explanation :

B is the correct answer.

append

Question - 56

```
class A extends Thread {  
    public void run() {  
        System.out.print("A");  
    }  
}  
class B {  
    public static void main (String[] args) {  
        A a = new A();  
        a.start();  
        a.start(); // 1  
    }  
}
```

- 1. compile time error
- 2. Runtime Exception
- 3. the code compile and runs fine

4.none of the above

Explanation :

B is the correct answer.

If the start method is invoked on a thread that is already running, then an `IllegalThreadStateException` will probably be thrown

Question - 57

```
class A extends Thread {
    public void run() {
        System.out.print("A");
    }
}
class B {
    public static void main (String[] args) {
        A a = new A();
        a.start();
        a.start(); // 1
    }
}
How many threads will be create ?
```

- 1.One
- 2.two
- 3.three
- 4.none of the above

Explanation :

A is the correct answer.

If the start method is invoked on a thread that is already running, then an `IllegalThreadStateException` will probably be thrown. One thread will be created.

Question - 58

```
class c1
{
    static{
        System.out.println("static");
    }
    public static void main(String a[])
    {
        System.out.println("main");
    }
}
```

- 1.prints static main
- 2.prints main

- 3.prints main static
- 4.compiletime error

Explanation :

A is the correct answer.

First execute the static block when the class is executed.

Question - 59

When a byte is added to a char, what is the type of the result?

- 1.byte
- 2.int
- 3.long
- 4.non of the above

Explanation :

B is the correct answer.

The result of all arithmetic performed with the binary operators (not the assignment operators) is an int, a long, a float, or a double. Here byte and char are promoted to int, so the result is an int.

Question - 60

Which of the following statements are true?

- 1) The automatic garbage collection of the JVM prevents programs from ever running out of memory
- 2) A program can suggest that garbage collection be performed but not force it
- 3) Garbage collection is platform independent
- 4) An object becomes eligible for garbage collection when all references denoting it are set to null.

- 1.1 and 4
- 2.2 and 4
- 3.2 and 3
- 4.3 and 4

Explanation :

B is the correct answer.

If a program keeps creating new references without any being discarded it may run out of memory. Unlike most aspects of Java garbage collection is platform dependent

